



MUST 'A' MATHS

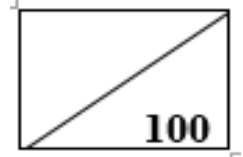
SM015

SESI 2024/2025

KOLEJ MATRIKULASI NEGERI SEMBILAN



Name: _____

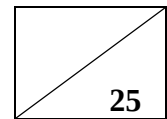


Class: _____

2 Hours

SECTION A [25 marks]

This section consists of 3 questions. Answer **all** questions.



1. Find the following limits, if they exist.

(a) $\lim_{x \rightarrow 3^-} \frac{|4x-12|}{x-3}$ [3 marks]

(b) $\lim_{x \rightarrow \infty} \frac{x^2 + x + 1}{(3x + 2)^2}$ [3 marks]

(c) $\lim_{x \rightarrow 8} \frac{8-x}{\sqrt{6}-\sqrt{x-2}}$ [3 marks]

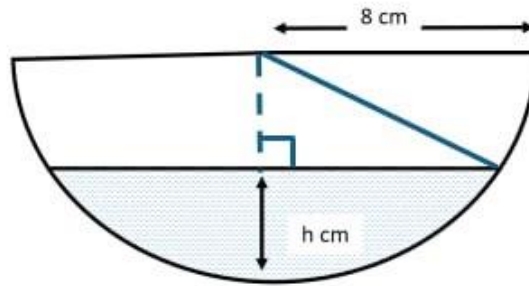
2. (a) Find $\frac{dy}{dx}$ for $y = 2 - x^2$ by using the first principle.

Hence, find the value of $\frac{dy}{dx}$ at $x = 2$. [4 marks]

(b) By using properties of logarithm, find differentiation of $k(x) = \ln\left(\frac{1 - \cos x}{\sin x}\right)$.

Give your answer in the simplest form.

[5 marks]



3. The above diagram shows a cross section of a hemispherical bowl of radius 8 cm.
- Water is poured into the bowl such that the height, h cm, of the water increases at a rate of 0.2 cm/s.

- (a) Show that the area of the surface of the water, $A \text{ cm}^2$ is given by

$$A = \pi(16h - h^2).$$

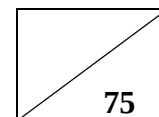
[3 marks]

- (b) Find the rate of increase of A when h is 6 cm.

[4 marks]

SECTION B [75 marks]

This section consists of 7 questions. Answer **all** questions.



1. Given that $(m + ni)(1 + i) = 1 + 2i$ and $|m| = \sqrt{3}$, where m is a complex number and n is a real number. Find all the possible m and all the values of n . [5 marks]

2. (a) Given $\frac{y+1}{y} \leq 2$. If $y = \log_2 x$, find the solution set of x . [7 marks]
(b) If $y = |1 - 2e^{-x}|$, solve $y = e^x$ for $1 - 2e^{-x} < 0$. [4 marks]

3. Consider $g(x) = \ln \sqrt{2x}$ and $f(x) = e^{k+2x}$, where k is a constant.
(a) Find $g^{-1}(x)$. Hence, show that $(g \circ g^{-1})(3) = 3$. [4 marks]
(b) If $(f \circ g)(x) = 2x$, find k . [4 marks]

4. Function g is defined as $g(x) = 2 - \frac{5}{x}$, $x \neq 0$.
(a) State the domain of $g(x)$. [1 marks]
(b) Show that $g^{-1}(x)$ exists. [2 marks]
(c) For the function $h(x) = \frac{5}{2-x}$, verify that $g(x)$ and $h(x)$ are inverse to each other.
Hence, find the domain and range of $h(x)$. [6 marks]
(d) On the same axes, sketch the graph of $g(x)$ and $g^{-1}(x)$. [3 marks]

5. (a) Find the horizontal and vertical asymptotes for $f(x) = \frac{\sqrt{x^2-1}}{x-1}$.

[5 marks]

- (b) A function $f(x)$ is defined by

$$f(x) = \begin{cases} \frac{x^2-1}{x-1} & , x < 1 \\ a^2-7 & , x = 1 \\ \frac{|1-x|}{x-1} + b & , x > 1 \end{cases}$$

Determine the values of the constants a and b if $f(x)$ is continuous at $x = 1$.

[9 marks]

6. (a) Find $\frac{dy}{dx}$ in terms of x and y , given that $\ln(3y) = y \ln x$, where $y > 0$ and $x > 0$.

Give your answer in the simplest form. Hence, evaluate $\frac{dy}{dx}$ when $x = 1$.

[7 marks]

- (b) Given $x = \cos 2t$, $y = \sin^2 t + 2e^{\ln t}$. Show that $\frac{dy}{dx} = -\frac{1}{2} - \operatorname{cosec} 2t$,

and evaluate $\frac{d^2y}{dx^2}$ when $t = \frac{\pi}{8}$.

[9 marks]

7. Given $f(x) = x^3 + 3x^2 - 9x + 6$.

- (a) By using the first derivative test, find the extremum points.

[4 marks]

- (b) Find the inflection points of $f(x)$.

[3 marks]

- (c) Sketch the graph.

[2 marks]

END OF QUESTION PAPER